

Solution Design

YAT Enrolline Cloud Architecture — Baseline Design

Engagement	YAT Enrolline Cloud Migration — Foundation Build
Document type	Technical design (Solution Design)
Version	v1.0 — Approved for implementation
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Approved by	Pat Lin (MTS Senior Consultant) · Sam Walker (YAT ICT Manager)
Status	As built — this is the deployed environment
Classification	Internal — YAT ICT, Student Administration, and MTS personnel on signed MSA

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1. Purpose and Scope

This document specifies the baseline AWS architecture MTS implemented as the foundation build phase of the YAT Enrolline cloud migration, for the student records and enrolment management system. The design stops at “infrastructure ready for application deployment”: the EC2 instance is provisioned with the OS, the RDS instance with an empty PostgreSQL engine, the S3 bucket empty, and the load balancer with placeholder health checks — no application binaries or student data were placed by the MTS build.

It is the **as-built** record of the current environment, and the starting point for any onward improvement of that environment.

In scope of this design

- The production cloud foundation for the Enrolline student records application.
- All compute, networking, identity, storage, database, autoscaling and baseline monitoring needed to run Enrolline as a staff-facing workload in AWS.
- An Amazon S3 document store for scanned student attachments, which are a working part of the record rather than an archive.
- Single-region, single-Availability-Zone deployment in ap-southeast-2 (Sydney).

Building this in a lab the design calls for ap-southeast-2 (Sydney). If you are building this in an AWS Academy Learner Lab, deploy to whichever region that environment gives you — us-east-1 — and treat it as standing in for Sydney throughout. That is acceptable.

Out of scope — deferred to the follow-on HA design phase

- High-availability hardening (Multi-AZ database, cross-AZ compute resilience, failure-simulation testing).
- Disaster recovery to a second AWS region; DR runbook and tabletop testing.
- Capacity scheduling against the academic calendar.
- Application re-platforming (Enrolline remains Amazon Linux + PostgreSQL).

Out of MTS scope entirely — YAT ICT responsibility

- Enrolline application installation onto the EC2 instance(s) after handover.
- Database migration (extract from the on-premises database, load into RDS for PostgreSQL).
- Migration of the scanned document library into the S3 store.
- Cutover — DNS switch, parallel running, decommissioning, user redirection (avoiding intake and census windows).
- Organisational change management — CAB approvals, communications, training, post-cutover support.

2. Design Inputs and Requirements

2.1 Inputs

- Enrolline Application Specification — functions, user load, data, integrations, SLAs, data residency.
- Enrolline Infrastructure Specifications — current operational state, utilisation, growth.
- Engagement Role Brief and Consultation Notes — engagement scope; OS, application and database preservation.
- YAT academic calendar — the intake enrolment windows and census dates that fix the Restricted Period.

2.2 Requirements the design must meet

Requirement	Target / note
Availability (business hours, outside intake)	99.5%

Availability (intake enrolment window)	99.9% expected by the business
RPO	≤ 1 hour
RTO	≤ 2 hours
Concurrency (typical / intake peak)	25–40 / 90–110
Data residency	Australian region only — Privacy Act 1988, APP 8
Student-record retention	30 years (NVR Standards / ASQA)
Application preservation	No change to Enrolline; PostgreSQL retained

3. Review of Existing Architecture

Enrolline previously ran on a single on-premises server co-locating the application and its database, with scanned attachments on a file share. Its single points of failure were the server itself, its storage, and the campus power and network feeding it. Recovery depended on nightly tape and was never tested against the intake calendar.

4. Architecture Design

4.1 Assumptions and constraints

#	Assumption / constraint	Source
A1	Enrolline runs unchanged on Amazon Linux 2023 with PostgreSQL	Role Brief
A2	Students do not access Enrolline; it is staff-facing only	Application Specification
A3	Intake and census dates are fixed externally and published in advance	Registrar
A4	Scanned attachments must be retained for 30 years	NVR Standards
C1	Single-AZ for the foundation build; HA deferred	Engagement scope
C2	All data remains in ap-southeast-2	Privacy / Data Handling Policy

4.2 AWS account and region

A dedicated AWS account in region ap-southeast-2 (Sydney). No resources are created outside that region.

4.3 Identity and Access Management (IAM)

Group	Purpose	Indicative permissions
EnrollineAdmins	YAT ICT administration of the environment	EC2, RDS, S3, CloudWatch — full within the account
EnrollineReadOnly	Audit and review access	ReadOnlyAccess
EnrollineAppRole	Instance profile for the application tier	Read/write to yat-enrolline-documents; CloudWatch Logs write

Building this in a lab the Learner Lab does not permit IAM user, group or role creation. Use the role the lab provides and note where these groups would sit in a real account.

4.4 Network topology

Subnet	CIDR	Tier	Internet-facing?
enrolline-public-a	10.30.1.0/24	Public (ALB, NAT)	Yes
enrolline-public-b	10.30.2.0/24	Public (ALB only)	Yes
enrolline-app-a	10.30.11.0/24	Application	No — egress via NAT
enrolline-data-a	10.30.21.0/24	Database	No
enrolline-data-b	10.30.22.0/24	Database (subnet group only)	No

VPC enrolline-vpc, 10.30.0.0/16. A single NAT Gateway in enrolline-public-a provides outbound internet for the application tier. There is no application subnet in the second Availability Zone. There are no VPC endpoints, so traffic to Amazon S3 and to the USI Registry web service leaves the VPC through the NAT Gateway.

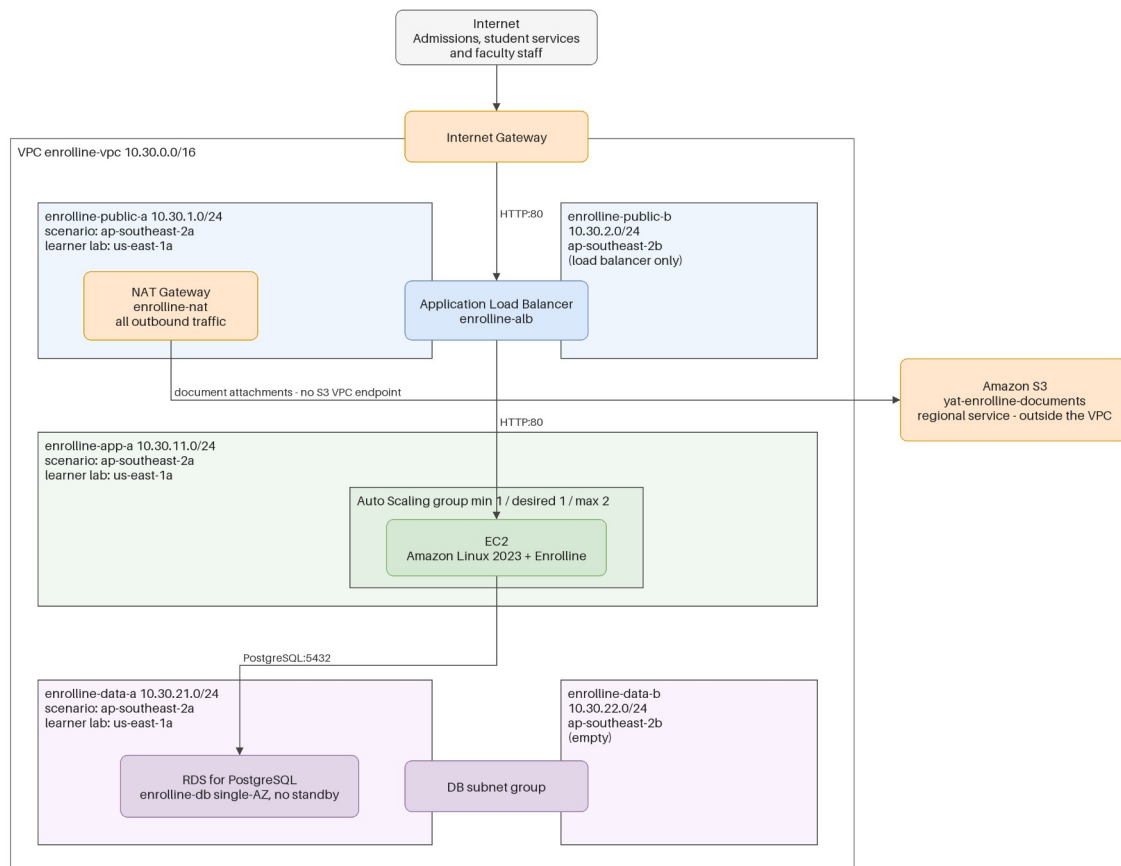


Figure 4.4 — Enrolline baseline network topology. The workload runs in ap-southeast-2a; the second-zone subnets carry only the load balancer and the database subnet group. Amazon S3 is a regional service and sits outside the VPC, reached through the NAT Gateway.

4.5 Compute (EC2 + Auto Scaling)

Attribute	Value
Instance family	General-purpose burstable (t3.micro / t3.small)

AMI	Amazon Linux 2023 + Enrolline Student Management Suite
Auto Scaling Group	min 1 / desired 1 / max 2 — enrolline-app-a only
Scaling policy	Target tracking on CPU at 70%
Capacity basis	Sized for the intake peak and held at that size year-round
Administrative access	Systems Manager Session Manager — no key pair, no bastion

4.6 Load balancing (ALB)

An internet-facing Application Load Balancer, enrolline-alb, spanning enrolline-public-a and enrolline-public-b, with an HTTP:80 listener to target group enrolline-tg. Health check HTTP GET on /, 30-second interval, 2 failures to remove a target. TLS is not terminated at this baseline.

4.7 Database (RDS for PostgreSQL)

Attribute	Value
Engine	Amazon RDS for PostgreSQL
Instance class	db.t3.micro / db.t3.small
Multi-AZ	Disabled — no standby instance
Storage	gp3, 20 GB; encrypted with AWS KMS
Placement	enrolline-data-a; not publicly accessible
Subnet group	enrolline-db-subnet-group, spanning both data subnets
Backup retention	7 days, automated, with transaction logs

4.8 Storage — the S3 document store

Attribute	Value
Bucket	yat-enrolline-documents
Contents	Scanned ID evidence, prior qualifications, USI evidence, support plans
Versioning	Enabled
Encryption	SSE-S3
Storage class	S3 Standard for all objects — no lifecycle configuration
Public access	Blocked at the bucket level
Access path	Via the NAT Gateway — no S3 VPC endpoint

4.9 Security

Security group	Inbound
enrolline-alb-sg	HTTP:80 from 0.0.0.0/0
enrolline-app-sg	HTTP:80 from enrolline-alb-sg only
enrolline-db-sg	PostgreSQL:5432 from enrolline-app-sg only

Encryption at rest is enabled on the database and the document store. No VPC flow logs, load-balancer access logs, or S3 access logging are configured at this baseline.

4.10 Monitoring (baseline)

- enrolline-unhealthy-hosts — any unhealthy target behind the load balancer.
- enrolline-db-storage-low — free database storage below 15% of allocation.

4.11 Naming and tagging conventions

All resources are prefixed enrolline- and tagged System=Enrolline, Environment=Production, Owner=YAT-ICT.

4.12 Backup

- Database: RDS automated daily backups, 7-day retention, plus transaction logs.
- Documents: S3 versioning. No cross-Region copy.

4.13 Recovery objectives — baseline state

The two-hour recovery-time objective is ****not reliably met**** by this baseline. Recovering a failed single-AZ database depends on point-in-time restore, which does not complete inside two hours at this data volume. This is a known limitation of the foundation build, recorded here rather than resolved.

4.14 Single points of failure

- RDS database — single instance, one Availability Zone, no standby.
- EC2 application — all capacity in enrolline-app-a, with no second-zone application subnet to expand into.
- NAT Gateway — single, in enrolline-public-a. Its loss also removes S3 document access and USI verification.

4.15 Configuration decisions left to the implementer

- Instance and database class within the stated families, on the documented load profile.
- CloudWatch alarm thresholds beyond the two baseline alarms.
- Tag values beyond the three mandatory keys.

5. Implementation Sequencing

- Network — VPC, subnets, gateways, routing, security groups.
- Storage — the S3 document store and its bucket policy.
- Database — subnet group, parameter group, RDS instance.
- Compute — launch template, Auto Scaling group, target group, load balancer.
- Monitoring — the two baseline alarms.

6. Verification Plan

- The load balancer reports a healthy target and serves the application endpoint.
- The application tier can read and write the document store.
- The database is reachable from the application tier and from nowhere else.
- Both baseline alarms exist and are in an OK state.

7. References

- Enrolline Application Specification (YAT ICT records)

- Enrolline Infrastructure Specifications (YAT ICT records)
- Enrolline Network Diagram (YAT ICT records)
- Enrolline Operational Costing (YAT ICT records)
- Backup and Retention Policy; Change Management Procedure; Privacy / Data Handling Policy (intranet)

Document control

Field	Value
Document owner	Pat Lin, MTS Senior Consultant
Business owner	Jess Tran, YAT Registrar
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Classification	Internal — YAT ICT, Student Administration, Compliance, and MTS personnel on signed MSA